ARA-290 (Cibinetide) Available for Research Use Only

(Repair 2 - 4 mg / day, Longevity M/W/F 4/2/1mg)

ARA-290 (Cibinetide) is an 11-amino-acid peptide derived from erythropoietin (EPO) that selectively activates the innate-repair receptor to produce tissue-protective, anti-inflammatory, and neuroregenerative signaling without stimulating erythropoiesis. It is used experimentally for small-fiber neuropathy and other nerve-injury syndromes.

💉 💉 Probably safe (some human evidence)

⭐ ⭐ Probably effective (some positive human signals)

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

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The evidence for ARA-290 long term human safety is limited because there are **no peer-reviewed long-term safety trials.** However, **all Phase II trials showed favorable safety profile,** with **no reported erythropoietic effects** and **generally good tolerability**. Across published trials, **no clinically meaningful increases in hemoglobin, hematocrit, or reticulocyte counts** were observed, supporting that ARA-290 **does not stimulate erythropoiesis** at therapeutic doses, a critical safety distinction from full-length EPO. The most commonly reported adverse events (AE) in trials were **mild and transient**, e.g., **headache, fatigue, and occasional injection-site reactions,** with low discontinuation rates attributable to AEs. Serious adverse events directly attributed to ARA-290 have not been reported in the Phase II datasets to date.

The evidence for ARA-290 efficacy is good, with clinical reports documenting **objective and symptomatic improvements**.

Dose Used In Trials

Human studies used several regimens, including **2 mg IV three times weekly and 4 mg SC daily for 28 days. Trials found improvements in small-fiber neuropathy symptoms, reduced pain scores, improved autonomic function, increased corneal nerve fiber density, and anti-inflammatory effects**. Human Trials at **4 mg/day produced stronger objective nerve-repair signal (corneal nerve fiber area and regenerating skin fibers).**

ARA-290 has a **short plasma half-life** but triggers sustained tissue responses when receptor engagement exceeds nanomolar affinity.

- **4 mg SC daily is the best-supported repair dose from human trials; other studied regimens include 2 mg IV three times weekly and 1–8 mg SC daily.**

Longevity / Healthspan Dose

In vitro research shows the **Innate Repair Receptor** has **high affinity** and responds to brief, low-level pulses. Even **very small extracellular concentrations of ARA-290 initiated the repair cascade**. Across multiple cell-based assays, IRR engagement and downstream signaling, including JAK2/STAT3 activation, NF-κB suppression, and macrophage M1→M2 polarization, occur in the **low-nanomolar range**, with measurable effects often appearing at **sub-nanomolar concentrations**.

Published human PK data are limited but include measured exposure after 4 mg SC and 2 mg IV.

Based on *in vitro* work:

- **Plausible receptor engagement** was found at **"sub- to low-nM."**
- **Plausible functional activation window** was found at **"~0.1 nM to a few tens of nM"** for endpoints like angiogenesis, anti-inflammatory gene programs.

The target tissue type should be taken into consideration when making dose decisions. For example, **disc endplate and nucleus pulposus** are relatively poorly perfused; extrapolating from non-ARA-290 PK, **interstitial concentrations** there will be lower than plasma and may require **2–10×** plasma overshoot to reach robust functional levels.

Lower intermittent doses might engage IRR signaling, but human PK/PD data do not establish a minimum effective dose for healthspan or chronic tissue protection. Extrapolating from mechanistic biology, plausible maintenance / longevity doses

- **0.5 mg SC overlaps the lower functional window; likely sufficient to *transiently* engage receptor signaling in well-perfused tissues and may trigger downstream signaling in some cell types.** For poorly perfused targets, this may be marginal after accounting for tissue access loss.
- **1 mg sits squarely within the lower–mid functional activation window; likely to produce a stronger and more reliable receptor engagement** signal systemically and in many tissues.
- **2.0 mg** provides a larger margin

Mechanistically, **all three doses plausibly reach the repair-cascade activation window**, with larger doses offering more certainty to account for incorrect assumptions in our mechanistically extrapolated PK data, and a stronger buffer against variability in tissue penetration.

The following table shows six kinds of tissue that might plausibly benefit from ARA-290 in the context of healthspan, their sensitivity / resistance to plasma concentration, and an extrapolation of the optimal application frequency based on the tissues' turnover, ongoing low-grade stress, etc. This extrapolation is speculative.

Tissue	Sensitivity to ARA-290 plasma concentration	Likely need for exposure frequency
Small-fiber peripheral nerves (sensory/autonomic)	**High**; IRR upregulation on stressed small fibers; effects seen at **sub- to low-nM** in vitro.	**Daily or every-other-day**; repeated pulses support ongoing repair and symptom control.
Microvascular endothelium	**High–moderate**; endothelial cytoprotection and anti-inflammatory signaling activated at **low-nM**.	**Every-other-day to weekly**; frequent pulses help maintain endothelial resilience but may tolerate spacing.
Immune cells / tissue macrophages	**Moderate**; macrophage M1→M2 shifts and cytokine suppression occur in **sub- to low-nM** ranges.	**Every-other-day**; periodic engagement reduces chronic inflammation while limiting desensitization.
Cardiac myocardium	**Moderate**; cardioprotective signaling observed in preclinical models at **low-nM** exposures.	**Every-other-day to weekly**; intermittent pulses likely sufficient for stress-resilience effects.
Central nervous system glia (microglia)	**Moderate–low**; anti-inflammatory effects reported at low-nM but CNS access is limited.	**Weekly or pulsed courses**; less frequent dosing may suffice due to slower CNS kinetics and tissue access limits.
Pancreatic islets and metabolic tissues	**Low–moderate**; preliminary signals for β-cell protection at low-nM but evidence is limited.	**Every-other-day to weekly**; conservative, intermittent exposure until stronger clinical data exist.

However, daily high dose ARA-290 for healthspan may not be practical: it is plausibly more likely to result in receptor desensitization, and it may be expensive. Taking into consideration that some tissues which benefit most from frequent dosing activate at lower plasma saturation levels, while several other tissues which need higher plasma concentration need less frequent dosing, the following is a hypothetical longevity dosing strategy.

- **4mg Monday, 2 mg Wednesday, 1 mg Friday for healthspan -** *Hypothetical regimen; not studied clinically or preclinically as a healthspan protocol*

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Speculatively, extrapolating this M/W/F 4/2/1 dose strategy to the six tissue types result in the following healthspan benefits while requiring just 28 mg for four weeks.

Tissue	Likely Activation	Plausible healthspan impact of 4–2–1 mg weekly dose
Small-fiber peripheral nerves	**M/W/F**	**Good nerve protection, moderate regeneration.** This pattern could meaningfully slow small-fiber loss, reduce neuropathic symptoms, and support autonomic stability.
Microvascular endothelium	**M/W/F**	**Solid microvascular resilience.** May improve endothelial health, capillary perfusion, and wound-healing capacity, supporting organ function and tissue repair with age.
Immune cells / macrophages	**M/W**	**Effective inflammaging modulation.** This schedule may lower chronic inflammatory burden, good for healthspan via reduced systemic inflammation.
Cardiac myocardium	**M/W**	**Only moderate cardio protection.** May enhance stress resilience and limits low-grade cardiac inflammation over time; plausibly helpful for preserving cardiac reserve in an aging context.
CNS glia / microglia	**M**	**Plausible modest neuroinflammatory control.** May reduce microglial overactivation and support cognitive health, especially over long horizons.
Pancreatic islets / metabolic tissues	**M**	**Supportive metabolic resilience.** Plausible modest improvements in stress tolerance and metabolic stability.

Benefits

- **Reduction in neuropathic pain and sensory symptoms**.
- **Objective small-fiber nerve regeneration**.
- **Protection and stabilization of small-fiber nerves**, helping slow nerve loss and support autonomic function over time.
- **Strong anti-inflammatory and tissue-protective signaling**, including suppression of pro-inflammatory cytokines, anti-apoptotic effects, and mitochondrial protection.
- **Plausible neuroinflammatory reduction**, supporting microglial stability and long-term cognitive health.
- **Improved microvascular resilience**, with better endothelial health, capillary perfusion, and wound-healing capacity.
- **Moderate cardioprotection**, enhancing stress resilience and reducing low-grade cardiac inflammation.

- **Supportive metabolic resilience**, with preliminary human evidence for improved glycemic and lipid markers.

Indications

- **Small-fiber neuropathy (sarcoidosis-associated)** — *Human clinical data*.
- **Diabetic small-fiber neuropathy / neuropathic symptoms** — *Human clinical data (small cohorts)*.
- **Experimental use for other nerve-injury or inflammatory neuropathies** — *Theoretical / animal evidence supports exploration*.

Contraindications

- **Concurrent use of agents that require erythropoietic stimulation monitoring** — *Theoretical / safety precaution* (although ARA-290 is nonerythropoietic, caution is advised in complex hematologic disease).
- **Uncontrolled cardiovascular disease where novel vasomodulatory agents are risky** — *Theoretical / precaution from trial exclusion criteria*.

Side Effects

- **Transient systemic symptoms (fatigue, headache)** — *Human clinical data*: mild, self-limited events reported in trials.
- **No erythropoiesis or hematocrit elevation reported in trials** — *Human clinical data* (important safety distinction).

ARA-290 and Healthspan

ARA-290's relevance to aging biology comes from its ability to **reduce chronic inflammation**, **protect small-fiber nerves**, and **preserve microvascular function,** three domains that decline predictably with age. IRR expression is preferentially induced/upregulated in stressed or inflamed cells, meaning **ARA-290 acts where damage is occurring**. This targeted activation leads to NF-κB suppression, reduced pro-inflammatory cytokines, and a shift of macrophages toward the M2 pro-repair phenotype, supporting healthier tissue environments over time.

In human studies, ARA-290 has demonstrated improvements in small-fiber neuropathy, autonomic function, and corneal nerve fiber density, suggesting a capacity to preserve or restore neuronal integrity. Because **small-fiber degeneration is strongly associated with aging, metabolic dysfunction, and systemic inflammation, these findings form a core part of the healthspan rationale**. Additional research shows ARA-290 can improve metabolic markers, protect pancreatic β-cells, and modulate microglial activation, linking it to broader domains of metabolic resilience and neuroinflammation control.

Animal studies further support ARA-290's role in aging biology: **chronic IRR activation is discussed for improvements in cardiac inflammation, and mitochondrial stress tolerance**, all of which are central hallmarks of aging. These findings suggest that periodic activation of the IRR pathway may help maintain tissue homeostasis and reduce age-related functional decline.

ARA-290 is plausible for healthspan extension because it provides targeted anti-inflammatory signaling, nerve protection, microvascular support, and metabolic resilience. Its mechanism aligns with modern longevity frameworks focused on reducing chronic inflammatory load, preserving neuronal function, and maintaining tissue repair capacity across the lifespan.

Drivers of Diminished Response and Mitigation Strategies

Receptor desensitization with repeated ARA-290 exposure has not been adequately studied; its likelihood is unknown, but mechanistically low-moderate because multiple papers describe ARA-290 as having

- **Rapid receptor engagement**
- **Short plasma half-life**
- **Long-lasting downstream anti-inflammatory and cytoprotective signaling**

This is the **opposite of a system prone to receptor desensitization**. The **risk for neutralizing antibodies is unstudied, but plausibly low-moderate** because published clinical studies of ARA-290 report acceptable safety and do not highlight neutralizing antibody formation. Chronic activation of JAK2, STAT5 and related survival and anti-inflammatory pathways can induce compensatory changes in gene expression and signaling thresholds, a phenomenon described in cytokine receptor biology, so **downstream pathway adaptation through intracellular attenuation is plausible with long term use**. Homeostatic mechanisms in the immune and vascular systems can counterbalance sustained reparative signaling with any continuous healthspan application, making **system level physiological adaptation** a concern.

No evidence-based cycling interval has been established. Mechanistically, cycling this low dose ARA-290 longevity protocol is plausible to mitigate waning. Specific cycling protocols and decisions should be viewed as experimental. A plausible, albeit arbitrary cycle suggestion avoiding plateauing effects and resulting in strong response when restarting might be

- **1 month off every 6–9 months**

Monitoring

In addition to standard baseline monitoring for CBC and metabolic panels, **monitoring hematology to confirm absence of erythropoiesis** is recommended.

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Applicability for Subjects with Cushing's Disease (Speculative)

ARA-290's **anti-inflammatory, pro-survival, and microvascular-protective biology** makes it a plausible candidate to address specific Cushing-related complications (small-fiber neuropathy, microvascular injury), but **no direct clinical evidence** supports its use in Cushing disease. ARA-290 targets the innate-repair receptor to trigger **anti-inflammatory, anti-apoptotic, and microvascular-protective signaling**; these actions **plausibly address several pathologies seen in Cushing disease.** Important gaps (immune dysregulation, metabolic milieu, and long-term safety) mean benefit is speculative and unproven.

Cushing disease can produce neuropathic symptoms, peripheral nerve abnormalities, muscle wasting, and impaired tissue repair. ARA-290 selectively activates the **innate-repair receptor (EPOR/CD131)** to induce **JAK/STAT and PI3K/Akt pro-survival signaling, suppress NF-κB–driven inflammation, and improve microvascular perfusion,** mechanisms that have reduced nerve inflammation and promoted small-fiber regeneration in human neuropathy trials and in animal models.

Potentially relevant Cushing-related targets ARA-290 addresses

- **Small-fiber neuropathy and neuropathic pain:** ARA-290 improved corneal nerve fiber density and pain in sarcoidosis/diabetic neuropathy trials, suggesting it can **promote small-fiber repair and reduce neuroinflammation**, which is relevant to Cushing-associated neuropathic symptoms.
- **Microvascular dysfunction and tissue perfusion:** By **stabilizing endothelium and improving microcirculation**, ARA-290 could mitigate ischemic contributors to tissue damage and impaired wound healing seen with hypercortisolism.
- **Inflammation and immune dysregulation:** ARA-290's **anti-inflammatory signaling** may counteract the paradoxical systemic inflammation and innate immune reprogramming reported in Cushing patients.

Important reasons ARA-290 might *not* work or could be limited for Cushing's

- **Different disease drivers:** Cushing disease's dominant driver is **chronic glucocorticoid excess**, which causes broad endocrine, metabolic, and catabolic effects (bone loss, muscle atrophy, hyperglycemia) that ARA-290's neurovascular focus does not directly reverse.
- **Immune complexity:** Cushing patients show **concurrent systemic inflammation and cellular immunosuppression**; modulating one axis (IRR signaling) may not correct the complex immune phenotype.

Practical considerations and safety

- **Evidence stage:** ARA-290 has Phase II data in neuropathy but **no Phase III confirmation** and is investigational.

- **Monitoring and interactions:** In Cushing patients (cardiometabolic risk, infection susceptibility), any experimental therapy requires specialist oversight; potential interactions with immunomodulation or vascular status must be considered.

Biological Mechanism

ARA-290 is an **EPO-derived 11-amino-acid peptide** engineered to selectively engage the **innate repair receptor (IRR)**, a heteromeric complex of **EPOR and CD131 (βcR)**, expressed on stressed **neurons, glia, endothelial, and immune cells.** Activation of the IRR triggers **JAK2/STAT and PI3K/Akt pro-survival signaling**, **suppresses NF-κB–mediated proinflammatory cytokine production**, and **stabilizes mitochondrial function**, collectively producing **anti-apoptotic, anti-inflammatory, microvascular-protective, and pro-regenerative effects**.

Crucially, ARA-290 is **engineered to avoid meaningful activation of the erythropoietic EPOR pathway** and has not produced erythropoiesis in reported studies, so it avoids red-cell stimulation and associated thrombotic risks observed with full-length EPO. Preclinical in vitro and animal models show **reduced neuronal apoptosis, improved nerve conduction and regeneration, and restored microvascular perfusion**, which aligns with the objective corneal nerve fiber improvements and symptom relief seen in small human trials.

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